

SEMESTER END / BACKLOG SUBJECT EXAMINATIONS - JULY / AUGUST 2025

Program	: B.E. – Information Science and Engineering	Semester	: VI
Course Name	: Machine Learning	Max. Marks	: 100
Course Code	: 21 / 22IS62	Duration	: 3 Hrs

Instructions to the Candidates:

- Answer one full question from each unit.

UNIT - I

- Briefly explain the process by which machine learning algorithms can be selected, applied, and evaluated for any given problem. CO1 (06)
 - Explain the difference between the following with an example for each CO1 (06)
 - Training, Testing and validation dataset
 - Over-fitting and under-fitting
 - Classification and regression ML algorithms.
 - In a Medical Classification Model to diagnose if a patient is diabetic or not 50,000 patients were examined. The number of TP = 8500, FP = 4500, TN = 35,500, FN = 1500. Do the following for this Medical Classification Model: CO1 (08)
 - Draw the Confusion Matrix
 - Calculate the Accuracy value
 - Assess the Sensitivity Value
 - Estimate the Specificity Value
 - Measure the Error Rate
 - Determine the Precision Value
 - Compute the Recall Value
 - Evaluate the F-Measure
- Differentiate supervised, unsupervised and semi-supervised machine learning approaches with an example for each. CO1 (06)
 - Explain and compare the following model evaluation techniques in machine learning: Holdout Method, Bootstrap, Random Sub- sampling, and Cross-Validation. CO1 (08)
 - Explain with an example the difference between the following: CO1 (06)
 - Bias and Variance of the ML Model
 - Variance and Covariance of the ML Model

UNIT - II

- Classify the following attributes as binary, discrete, or continuous. Also classify them as qualitative (nominal or ordinal) or quantitative (interval or ratio). CO2 (07)
 - Time in terms of AM or PM.
 - Brightness as measured by a light meter.
 - Brightness as measured by people's judgments.
 - Angles as measured in degrees between 0° and 360°.
 - Apply Candidate – Elimination algorithm to solve the following data. CO2 (08)

Sky	Temp	Humidity	Wind	Water	Forecast	Enjoys
Sunny	Warm	Normal	Strong	Warm	Same	Yes
Sunny	Warm	High	Strong	Warm	Same	Yes
Rainy	Cold	High	Strong	Warm	Change	No
Sunny	Warm	High	Strong	Cool	Change	Yes
 - Explain the terms within-class scatter matrix and between-class scatter matrix used in LDA. Why are they important? CO2 (05)

4. a) Calculate the dissimilarity measure for asymmetric binary attributes. CO2 (07)

Name	Fever	Cough	Test 1	Test 2	Test 3	Test 4
Asad	Negative	Yes	Negative	Positive	Negative	Negative
Bilal	Negative	Yes	Negative	Positive	Positive	Negative
Tahir	Positive	Yes	Negative	Negative	Negative	Negative

- b) Apply Find -S algorithm for the following example to determine the final hypothesis. CO2 (07)

Weather	Temperature	Play
Sunny	Warm	Yes
Sunny	Cold	Yes
Rainy	Cold	No
Sunny	Warm	Yes

- c) Discuss how eigenvalues and eigenvectors are used to find the projection matrix in LDA. CO2 (06)

UNIT - III

5. a) The values of x and their corresponding values of y are shown in the table below. CO3 (08)

x	0	1	2	3	4
y	2	3	5	4	6

- i) Find the least square regression line $y = a x + b$.
 ii) Estimate the value of y when $x = 10$.
 b) For the dataset given below, apply CART Algorithm to draw the Final decision tree. CO3 (12)

Day	Outlook	Temp	Humidity	Wind	Play Tennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

6. a) Apply naïve Bayes algorithm for the given dataset and estimate the conditional probability of each attribute for the species class {M, H}. Identify the new instance {Color=Green, Legs=2, Height=Tall, smelly=No} belongs to which species CO3 (10)

No	Color	Legs	Height	Smelly	Species
1	White	3	Short	Yes	M
2	Green	2	Tall	No	M
3	Green	3	Short	Yes	M
4	White	3	Short	Yes	M
5	Green	2	Short	No	H
6	White	2	Tall	No	H
7	White	2	Tall	No	H
8	White	2	Short	Yes	H

- b) Suppose that the following are a set of points in two classes:

CO3 (10)

Positive Labeled Points	Negatively Labeled Points
$\left\{\begin{pmatrix} 3 \\ 1 \end{pmatrix}, \begin{pmatrix} 3 \\ -1 \end{pmatrix}, \begin{pmatrix} 6 \\ 1 \end{pmatrix}, \begin{pmatrix} 6 \\ -1 \end{pmatrix}\right\}$	$\left\{\begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ -1 \end{pmatrix}, \begin{pmatrix} -1 \\ 0 \end{pmatrix}\right\}$

- Plot the above points.
- Is the data linearly separable? Justify
- Which are the Support Vectors with and without bias?
- Find the separating hyper-plane that separates the positive and negative classes
- Update the plot to depict this hyper plane.

UNIT- IV

7. a) Group the following points into K=2 clusters using k-means algorithm and let the initial centroids be (2,3) and (5,6)- (2 iterations) CO4 (07)

Points	1	2	3	4	5	6	7	8
X	2	5	8	1	2	6	3	8
Y	3	6	7	4	2	7	4	6

- Differentiate between 'AGNES' and 'DIANA' as Hierarchical Clustering techniques. CO4 (05)
- Given below is the proximity matrix for 5 points. Show the complete link clustering and dendrogram for the following. CO3 (08)

	P1	P2	P3	P4	P5
P1	0.00	0.15	0.20	0.14	0.25
P2	0.15	0.00	0.15	0.28	0.11
P3	0.20	0.15	0.00	0.29	0.22
P4	0.14	0.28	0.29	0.00	0.39
P5	0.25	0.11	0.22	0.39	0.00

8. a) Using the single-linkage agglomerative clustering technique, cluster the data points for the below given distance matrix and design a Dendrogram. CO4 (08)

	a	b	c	d	e
a	0	9	3	6	11
b	9	0	7	5	10
c	3	7	0	9	2
d	6	5	9	0	8
e	11	10	2	8	0

- How does DBSCAN differentiate between core points, border points, and noise? CO4 (06)
- What are the similarities and differences between average-link clustering and k-means? CO4 (06)

UNIT - V

- Describe the concept of neural firing threshold and its analogy in artificial neurons. CO5 (07)
 - Name and briefly describe three methods of feature subset selection. CO5 (06)
 - How does boosting differ from bagging in its approach to combining models? CO5 (07)

10. a) Explain how bagging can improve the performance of unstable learners such as decision trees. CO5 (06)
- b) Compare and contrast dendrites and axons in terms of their roles in neural communication. CO5 (07)
- c) Given a dataset with 2 variables, how do you manually compute the principal components? CO5 (07)
